Skills Summary

Equivalent Expressions: Expanded and Factored Forms

Use this reference while completing your worksheet or when studying.

One property, two directions.

$$a(b + c) \;=\; ab + ac$$

Left to right it **expands**; right to left it **factors**. Two expressions are **equivalent** when they agree for *every* value of the variable — so one disagreeing substitution disproves equivalence, while one agreeing substitution proves nothing.

1. Expanding with the distributive property

The multiplier outside touches **every** term inside — one product per term. Write the inside as a sum of signed terms ($x - 7$ is $x + (-7)$) so the sign rules do the work: negative times negative is positive.

Example: Expand $4(x + 3)$.
Step 1. The parentheses hold two terms, so there will be two products — write them before simplifying anything.
Step 2. $4 \cdot x = 4x$ and $4 \cdot 3 = 12$, so the sum of the products is $4x + 12$.

$$4(x + 3) = \mathbf{4x + 12}$$

Try it: Expand $5(2x + 7)$.

check: $\mathbf{10x + 35}$

2. Factoring out the greatest common factor

Break each term into prime factors, take everything the terms *share*, and write it outside. Divide each original term by that factor to get what stays inside.

Completely means the terms left inside share nothing: if they still do, you pulled out a common factor, not the *greatest* one.

Example: Factor $6x + 9$ completely.
Step 1. Factoring is expanding read backwards, so start from what the two terms have in common rather than from the expression as a whole.
Step 2. $6x = 2 \cdot 3 \cdot x$ and $9 = 3 \cdot 3$ share only 3, and dividing gives $2x$ and 3 inside.

$$6x + 9 = \mathbf{3(2x + 3)}$$

Try it: Factor $14x - 21$ completely.

check: $\mathbf{7(2x - 3)}$

3. Collecting like terms

Like terms have the **identical variable part**: $5x$ and $2x$ are like, $5x$ and $5y$ are not, and $5x$ and 9 are not. Add or subtract only the coefficients; the variable part never changes.

If parentheses are present, expand first — nothing can be collected through a set of parentheses.

Example: Write $5x + 2x + 9$ with as few terms as possible.
Step 1. Sort the terms by their variable part first, so that only genuinely like terms get added.
Step 2. $5x$ and $2x$ are like and give $7x$; the 9 has no x, so it is carried along unchanged.

$$5x + 2x + 9 = \mathbf{7x + 9}$$

Try it: Write $3(x + 2) + 2x$ with as few terms as possible.

check: $\mathbf{5x + 6}$

4. Testing equivalence by substitution

Pick a value, substitute it into **both** expressions, and compare.

- Different values $\Rightarrow$ **not equivalent**. One counterexample is a complete disproof.
- Same value $\Rightarrow$ evidence only. Two different expressions can still cross at one point.

Avoid $x = 0$ and $x = 1$ as test values — too many wrong expressions agree there by accident.

Example: Find the value of $3(x + 5)$ at $x = 4$, to test it against a claimed expansion.
Step 1. Evaluate the correct expression first, so there is a reference value to compare a claim against.
Step 2. Substituting gives $3(4 + 5) = 3 \cdot 9$, and that product is 27.

$$3(x + 5) \text{ at } x = 4 \;=\; \mathbf{27}$$

Try it: Find the value of $2(x - 3)$ at $x = 7$.

check: **8**

(!) Watch out: distribute to *every* term, not just the first — $3(x + 4)$ is $3x + 12$, never $3x + 4$. When the multiplier is negative, both signs inside change. And after factoring, check that the terms inside the parentheses no longer share a factor.